

P11: Vectors (Year 1)

1. Vector Notation and Basics

Vector vs Scalar:

A **scalar** has magnitude only (e.g., speed, temperature).

A **vector** has both magnitude and direction (e.g., velocity, displacement, force).

Notation:

- Bold letters: **a** (printed) or a (handwritten)
- Arrow notation: $\vec{AB}$ — vector from A to B
- Column vector: $\begin{pmatrix} x \\ y \end{pmatrix}$ — x-component, y-component
- Component form: $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$, where $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Two vectors are **equal** if they have the same magnitude and direction — position does not matter (free vectors).

2. Vector Operations

Addition and Subtraction

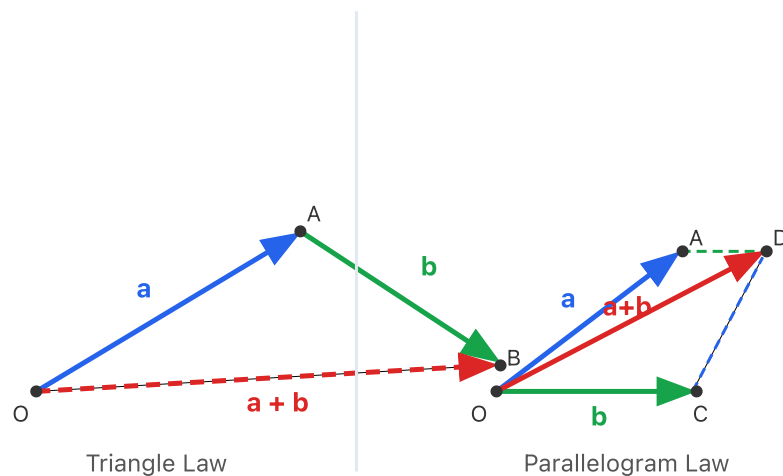
$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix} \quad \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}$$

Scalar Multiplication

$$k \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} ka_1 \\ ka_2 \end{pmatrix}$$

Multiplying by a positive scalar: scales the magnitude, keeps direction.

Multiplying by a negative scalar: scales and **reverses** direction.



Example — Given $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j}$, find:

(a) $\mathbf{a} + \mathbf{b} = (3 - 1)\mathbf{i} + (2 + 4)\mathbf{j} = 2\mathbf{i} + 6\mathbf{j}$

(b) $\mathbf{a} - \mathbf{b} = (3 + 1)\mathbf{i} + (2 - 4)\mathbf{j} = 4\mathbf{i} - 2\mathbf{j}$

(c) $3\mathbf{a} - 2\mathbf{b} = (9 + 2)\mathbf{i} + (6 - 8)\mathbf{j} = 11\mathbf{i} - 2\mathbf{j}$

3. Magnitude and Unit Vectors

Magnitude (length) of a vector:

$$|\mathbf{a}| = \left| \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \right| = \sqrt{a_1^2 + a_2^2}$$

This is just Pythagoras' theorem applied to the components.

Unit vector: A vector of magnitude 1 in the direction of $\mathbf{a}$:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Example — Find the unit vector in the direction of $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$:

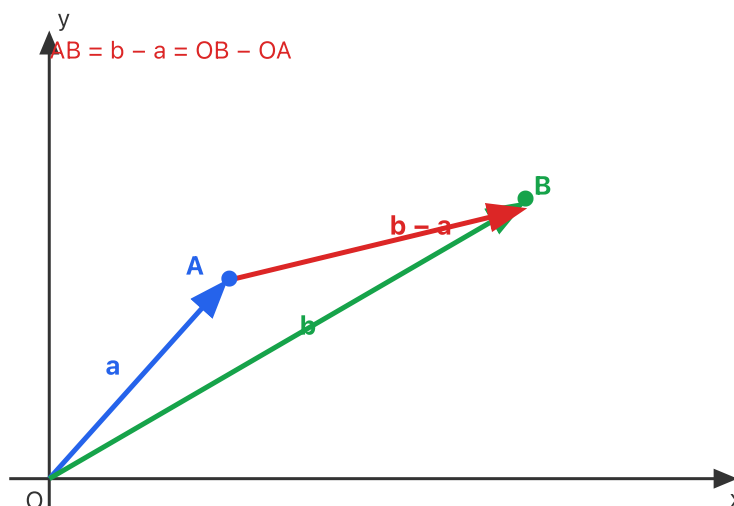
$$|\mathbf{a}| = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\hat{\mathbf{a}} = \frac{1}{5}(3\mathbf{i} + 4\mathbf{j}) = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$$

$$\text{Check: } |\hat{\mathbf{a}}| = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{1} = 1 \checkmark$$

Exam Tip: $\mathbf{i}$ and $\mathbf{j}$ are unit vectors. Note: $|\mathbf{i}| = |\mathbf{j}| = 1$, and they are perpendicular to each other.

4. Position Vectors and $\vec{AB}$



Position Vectors:

The position vector of a point P is $\vec{OP} = \mathbf{p}$ — the vector from the origin O to P.

Vector between two points:

$$\vec{AB} = \vec{OB} - \vec{OA} = \mathbf{b} - \mathbf{a}$$

This is the vector "going from A to B" — always: destination minus start.

Example — A has position vector $\mathbf{a} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, B has position vector $\mathbf{b} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$:

$$\vec{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 6 - 2 \\ 1 - 5 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \end{pmatrix}$$

$$|\vec{AB}| = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

Midpoint M of AB has position vector: $\frac{\mathbf{a} + \mathbf{b}}{2} = \frac{1}{2} \begin{pmatrix} 8 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

5. Midpoints and Dividing a Line in a Ratio

Midpoint: The midpoint M of AB has position vector $\frac{\mathbf{a} + \mathbf{b}}{2}$.

Dividing in ratio $m : n$: Point P dividing AB in ratio $m : n$ has position vector:

$$\vec{OP} = \frac{n\mathbf{a} + m\mathbf{b}}{m + n}$$

Example — Find the position vector of P that divides AB in the ratio 2:3, where $\mathbf{a} = \mathbf{i} + 4\mathbf{j}$ and $\mathbf{b} = 6\mathbf{i} - \mathbf{j}$:

$$\vec{OP} = \frac{3\mathbf{a} + 2\mathbf{b}}{5} = \frac{3(\mathbf{i} + 4\mathbf{j}) + 2(6\mathbf{i} - \mathbf{j})}{5} = \frac{(3 + 12)\mathbf{i} + (12 - 2)\mathbf{j}}{5} = \frac{15\mathbf{i} + 10\mathbf{j}}{5} = 3\mathbf{i} + 2\mathbf{j}$$

6. Proving Geometric Results with Vectors

Parallel Vectors

Two vectors $\mathbf{p}$ and $\mathbf{q}$ are **parallel** if $\mathbf{p} = k\mathbf{q}$ for some scalar k .

If $k > 0$: same direction. If $k < 0$: opposite direction.

Collinear Points

Points A, B, C are **collinear** (on the same line) if $\vec{AB} = k\vec{AC}$ for some scalar k .

This means $\vec{AB}$ and $\vec{AC}$ are parallel and share point A.

Example — Show that A(1,2), B(3,6), C(5,10) are collinear:

$$\vec{AB} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \vec{AC} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\vec{AB}$$

So $\vec{AC} = 2\vec{AB}$ — parallel and share point A — therefore A, B, C are collinear.

Example (geometric proof) — OABC is a parallelogram with $\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$. M is the midpoint of AB and N is the midpoint of CB. Show that MN is parallel to AC.

$$\vec{OB} = \mathbf{a} + \mathbf{c} \text{ (diagonal of parallelogram)}$$

$$\text{M is midpoint of AB: } \vec{OM} = \mathbf{a} + \frac{1}{2}\mathbf{c}$$

$$\text{Midpoint N of CB: } \vec{ON} = \frac{1}{2}(\vec{OC} + \vec{OB}) = \frac{1}{2}(\mathbf{c} + \mathbf{a} + \mathbf{c}) = \frac{1}{2}\mathbf{a} + \mathbf{c}$$

$$\vec{MN} = \vec{ON} - \vec{OM} = \frac{1}{2}\mathbf{c} - \frac{1}{2}\mathbf{a} = \frac{1}{2}(\mathbf{c} - \mathbf{a}). \text{ Since } \vec{AC} = \mathbf{c} - \mathbf{a}, \vec{MN} = \frac{1}{2}\vec{AC}, \text{ so MN is parallel to AC.}$$

7. Summary Table

Concept	Formula
Magnitude in 2D	$ \mathbf{a} = \sqrt{a_1^2 + a_2^2}$

Unit vector	$\hat{\mathbf{a}} = \frac{\mathbf{a}}{ \mathbf{a} }$
Vector A to B	$\vec{AB} = \mathbf{b} - \mathbf{a}$
Midpoint of AB	$\frac{\mathbf{a} + \mathbf{b}}{2}$
Divide AB in ratio m:n	$\frac{n\mathbf{a} + m\mathbf{b}}{m + n}$
Parallel condition	$\mathbf{p} = k\mathbf{q}$ for some scalar k
Collinear A, B, C	$\vec{AB} = k\vec{AC}$ for some scalar k

Key Exam Points for P11:

- Always write $\vec{AB} = \mathbf{b} - \mathbf{a}$ (destination minus start — not the other way round)
- Magnitude: use Pythagoras on components; simplify surds in your answer
- For collinearity proofs: show two vectors through a common point are parallel ($\mathbf{p} = k\mathbf{q}$)
- For geometric proofs: set up all position vectors from origin O, express each point in terms of the given base vectors
- i and j components must be equal independently when setting vector equations — equate coefficients of i and j separately